

Product Lifecycle

Smart Cold-Chain System — Farm-to-Market Closed Cold-Chain

The complete lifecycle is presented as a Farm-to-Market Closed Cold-Chain: produce is protected from harvest until sale while the system continuously tracks quality, temperature, humidity, battery status, location, and delivery status.

1. Harvest at Farm

Fresh fruits/vegetables are harvested at the appropriate maturity stage. The farmer selects and grades produce, removes damaged/spoiled items, and enters farmer ID, crop/product, quantity, harvest date and destination market into the system. A Crate ID is assigned.

- Output: fresh, sorted produce ready for cold-chain entry.

2. Collection & Pre-Processing

Produce moves from farm to the nearby collection/cold-storage facility, where it is inspected, quality and quantity are recorded, and it is placed into standardized reusable crates linked to the farmer and product information.

- Traceability chain established: Farmer → Product → Crate ID → Storage Location → Transport → Market.

3. Central Cold-Storage Entry

Produce enters the centralized modular cold-storage facility (grid-based storage blocks, each crate assigned a position). The system records Crate ID, grid/slot number, product type, quantity, entry time, required temperature/humidity range and farmer information.

4. Centralized Refrigeration & Storage

Grid power operates central refrigeration for the warehouse; solar power is used primarily to charge individual crate batteries while stored. Sensors continuously monitor temperature, humidity, power status and storage condition; the infra dashboard shows slots, solar generation and crate status.

5. Crate Preparation for Transportation

When a shipment is scheduled, the crate is identified by its Unique Crate ID. Before dispatch, the system verifies: produce already at required temperature, sufficient battery SOC, estimated temperature-maintenance time greater than expected journey duration, and functioning sensors/comms.

- The transport crate is not designed to rapidly cool warm produce — only to maintain the existing condition.

6. Convert Storage Crate → Intelligent Transport Crate

The same crate is lifted from its grid slot directly into the vehicle — no produce transfer. Each crate carries PU/PIR insulation, an SHT31 sensor, ESP32-C3 controller, TEC/Peltier actuator, LiFePO4 battery + BMS,

LCD/OLED + LED/buzzer display, and a tracking/communication module.

7. Intelligent Loading for Hilly Terrain

For remote/hilly Northeast routes, the system adapts strategy: route condition determines vehicle selection and crate count per trip. Difficult routes use lighter vehicles with fewer crates, and the system verifies battery runtime is sufficient for the expected travel time.

8. Transportation with Closed-Loop Temperature Maintenance

Once moving, the crate operates independently: SHT31 → ESP32 → temperature comparison → Peltier control → feedback. Within range, the Peltier stays at minimum/off; rising temperature activates the actuator; as it approaches setpoint, duty cycle is reduced — minimizing battery use.

9. Real-Time Monitoring During Transportation

The system continuously streams Crate ID, location, temperature, humidity, battery SOC, power consumption, estimated runtime and transport status to the backend. Farmers see their own crate; the infra team monitors the full fleet.

10. Predictive Battery & Temperature-Maintenance Estimation

Rather than showing raw battery %, the system calculates estimated time remaining to maintain the required temperature, using SOC, power draw, ambient temperature, setpoint, Peltier duty cycle, insulation characteristics and historical consumption — then compares this against remaining journey time, raising a critical alert if runtime falls short.

11. Automated Alerts

Abnormal conditions trigger immediate notification: temperature (high/low/rapid rise), humidity (high/low), battery (low SOC, abnormal consumption, insufficient runtime), and device faults (sensor failure, communication loss, controller/actuator fault). Farmers get crate-specific alerts; management gets system-wide alerts.

12. Arrival at Market

GPS/location, Crate ID and delivery confirmation mark the shipment delivered. The farmer is notified their crate arrived safely; the receiving point confirms quantity, product condition, delivery time and temperature history. The complete journey history is stored.

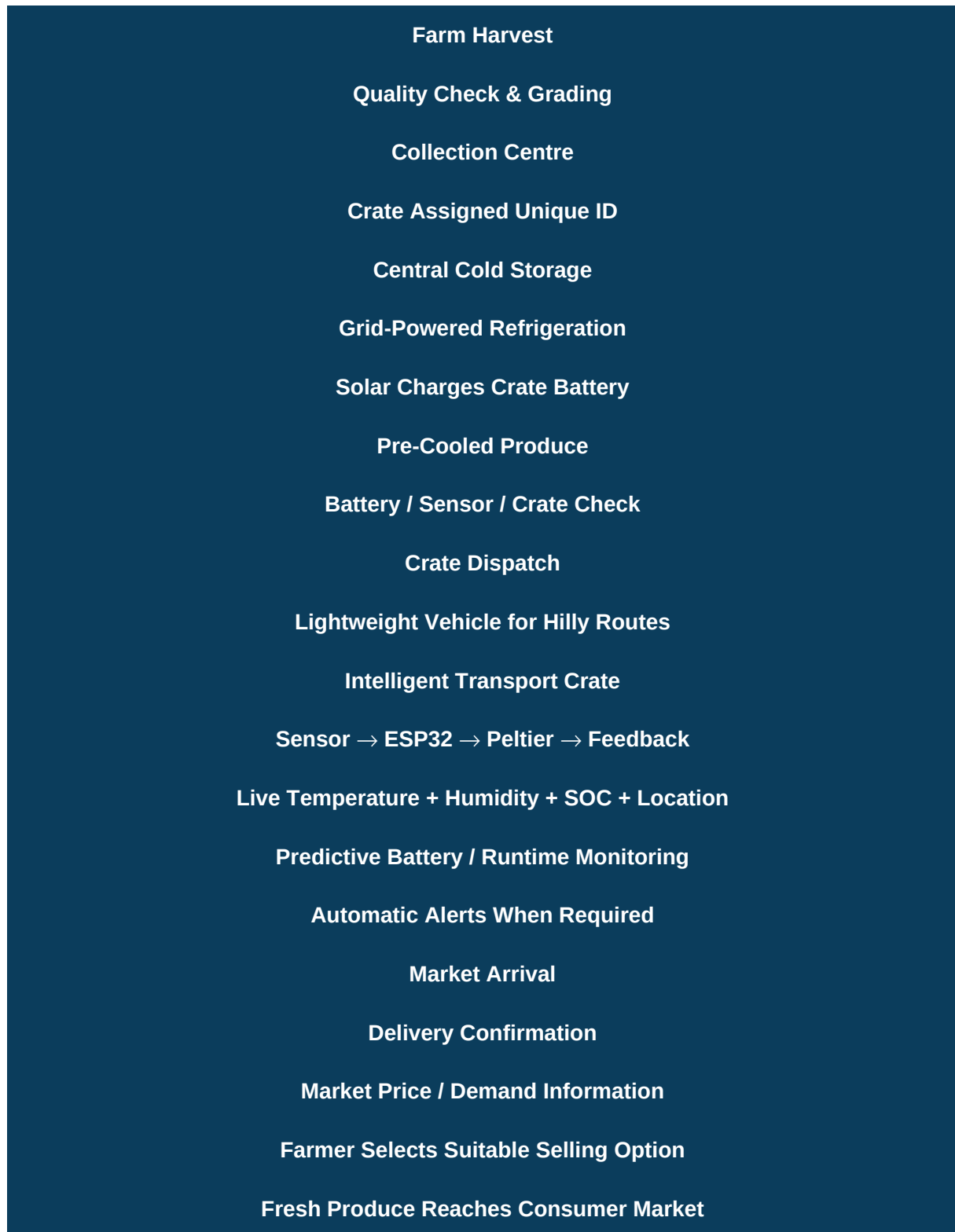
13. Market Readiness & Best-Price Decision

The farmer dashboard shows destination market, current/expected price, demand information, recommended selling location, estimated revenue, product condition and delivery confirmation — presented as a recommendation, not a guaranteed best price, since market conditions can change.

14. Sale of Fresh Produce

The final objective: fresh produce + traceability + timely delivery + market information → a better selling opportunity, with the farmer selling while produce remains within the desired quality window.

End-to-End Lifecycle Flow



Dashboard Summary

Infrastructure Management Dashboard — manage the entire cold-chain infrastructure (facility overview, energy management, crate management, transportation management, alerts, analytics).

Farmer Dashboard — track my crate, protect my produce, and know when/where to sell (my crate, live condition, tracking, notifications, market/delivery information, history).

Strongest farmer-facing feature chain for presentation: Unique Crate ID → Live Location → Temperature/Humidity → Battery SOC → Estimated Remaining Runtime → Delivery Confirmation → Best Selling Price.

Power Scenario Operating Modes

Scenario	Refrigeration	Crate Battery Charging	Main Strategy
Grid OFF + Solar ON	Solar emergency supply	Solar	Solar-Assisted Emergency Operation
Grid ON + Solar OFF	Grid	Grid charger	Normal Grid Operation
Grid OFF + Solar OFF	Thermal preservation only	No charging	PCM + Insulation + Priority Dispatch (Thermal Survival Mode)

Case 3 should be presented as a 'Thermal Survival Mode' — the goal is to maximize safe holding time and prevent sudden spoilage until power returns or produce is dispatched, not to run refrigeration indefinitely without power.